

Coffee Shop Sales Analysis

Business Intelligence Report Using Microsoft Excel

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Executive Summary

In today's competitive retail environment, businesses rely on data-driven decision-making to improve operational efficiency, increase profitability, and enhance customer satisfaction. Coffee shops generate a significant volume of transactional data every day, making it essential to analyze sales trends, customer purchasing behavior, and product performance. Without proper analysis, valuable business opportunities may remain undiscovered.

This project presents a comprehensive sales analysis of a coffee shop using Microsoft Excel. The analysis was performed on transactional sales data collected over six months, covering January through June. The dataset includes information related to transaction dates and times, store locations, product categories, product types, quantities sold, unit prices, product sizes, and total sales amounts.

The primary objective of this project was to transform raw sales data into meaningful business insights through data cleaning, transformation, Pivot Tables, and an interactive dashboard. Microsoft Excel was used as the sole analytical tool to organize the data, calculate key performance indicators (KPIs), and develop dynamic visualizations that enable users to explore sales performance across different dimensions.

The dashboard reveals several important insights regarding business performance. The coffee shop generated a total sales revenue of **\$698,812.33** from **149,116 orders**, with an average order value of **\$4.69** and an average of **1.44 items purchased per transaction**. Coffee products emerged as the highest revenue-generating category, accounting for **39% of total sales**, while Barista Espresso proved to be the highest-performing product type. Sales activity peaked during the morning hours, particularly between **8:00 AM and 10:00 AM**, indicating strong demand during breakfast and commuting hours. Among the three store locations, Hell's Kitchen recorded the highest revenue, although all stores demonstrated relatively balanced performance.

The findings of this analysis provide valuable recommendations for business improvement, including optimizing staffing during peak hours, promoting top-performing products, improving data quality, enhancing product bundling strategies, and strengthening customer engagement through targeted promotions.

Overall, this project demonstrates how Microsoft Excel can effectively support business intelligence and decision-making by converting raw transactional data into actionable insights through structured analysis and interactive visualizations.

1. Introduction

The retail food and beverage industry is highly competitive, requiring businesses to continuously monitor their sales performance and customer purchasing behavior. Coffee shops, in particular, serve hundreds or even thousands of customers each day, generating large volumes of transactional data. While this data contains valuable information, it cannot contribute to business growth unless it is properly organized, analyzed, and interpreted.

Sales analysis enables businesses to understand customer preferences, identify popular products, recognize peak business hours, compare store performance, and evaluate overall revenue generation. These insights help management make informed decisions regarding inventory management, staffing, pricing strategies, promotional campaigns, and future business planning.

With the increasing availability of business data, Microsoft Excel remains one of the most widely used analytical tools due to its powerful capabilities in data cleaning, transformation, visualization, and reporting. Features such as Pivot Tables, Pivot Charts, slicers, formulas, and dashboards allow analysts to summarize complex datasets and present findings in a clear and interactive manner.

This project focuses on analyzing retail sales data from a coffee shop using Microsoft Excel. The objective is not only to summarize historical sales performance but also to identify patterns and trends that can improve operational efficiency and support strategic business decisions.

An interactive dashboard was developed to visualize important performance indicators, allowing users to filter data by month and day while exploring sales across multiple dimensions such as time, product category, store location, and product type. Through this dashboard, managers can quickly identify business opportunities and respond effectively to changing customer demand.

2. Problem Statement

Coffee shops process thousands of customer transactions over time, generating large amounts of sales data. Although this information contains valuable insights, manually reviewing thousands of records is both time-consuming and inefficient. Without systematic analysis, it becomes difficult for management to identify customer purchasing patterns, evaluate store performance, monitor sales trends, or determine which products contribute most to business revenue.

The absence of an analytical reporting system can lead to several operational challenges, including ineffective staffing during peak business hours, poor inventory planning, missed sales opportunities, and limited understanding of customer preferences. As a result, business decisions may rely on assumptions rather than factual evidence.

To address these challenges, this project develops an interactive sales analysis dashboard using Microsoft Excel. By cleaning, transforming, and visualizing transactional sales data, the dashboard enables management to monitor key performance indicators, analyze sales across multiple dimensions, and make informed business decisions based on accurate and timely information.

3. Project Objectives

The primary objective of this project is to analyze coffee shop retail sales data and generate meaningful business insights that support effective decision-making. The analysis aims to answer important business questions related to sales performance, customer purchasing behavior, product popularity, and operational efficiency.

The specific objectives of this project are:

- To analyze how sales vary by day of the week and hour of the day.
 - To identify peak business hours with the highest customer activity.
 - To calculate and compare monthly sales revenue.
 - To evaluate sales performance across different store locations.
 - To determine the average order value and average number of items purchased per transaction.
 - To identify the best-selling products based on revenue generation.
 - To analyze revenue distribution across different product categories and product types.
 - To design an interactive dashboard that enables users to explore sales trends dynamically using filters and slicers.
 - To provide practical business recommendations based on analytical findings.
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4. Dataset Description

The dataset used in this project contains transactional sales records collected from multiple coffee shop locations over a six-month period. Each record represents an individual customer transaction and includes information related to products, stores, quantities, pricing, and transaction timing.

The dataset consists of the following attributes:

Column Name	Description
Transaction ID	Unique identifier assigned to each transaction.
Transaction Date	Date on which the transaction occurred.
Transaction Time	Time at which the purchase was completed.
Store ID	Unique identifier for each coffee shop branch.
Store Location	Name of the store where the transaction took place.
Product ID	Unique identifier assigned to each product.
Transaction Quantity	Number of units purchased in a transaction.
Unit Price	Selling price of a single product unit.
Product Category	General classification of products such as Coffee, Bakery, or Tea.
Product Type	Specific type of product within each category.
Product Detail	Detailed description of the product.
Size	Beverage or product size purchased by the customer.
Total Amount	Total sales value generated from the transaction.
Month Name	Name of the month extracted from the transaction date.
Day Name	Name of the day extracted from the transaction date.
Hour	Hour extracted from the transaction time for time-based analysis.
Month	Numerical representation of the month.
Day of Week	Numerical representation of the weekday for analysis purposes.

5. Data Cleaning and Transformation

Raw data often contains inconsistencies that can affect the accuracy of analysis. Before creating the dashboard, the dataset was cleaned and transformed in Microsoft Excel to ensure reliable and meaningful results.

The data preparation process included the following steps:

5.1 Data Validation

The dataset was reviewed to ensure that each transaction contained complete and consistent information. Numerical fields such as **Transaction Quantity**, **Unit Price**, and **Total Amount** were checked to verify that they contained valid values suitable for analysis.

5.2 Formatting

Columns containing dates and times were formatted appropriately to enable time-based analysis. This allowed the extraction of additional information such as months, weekdays, and hourly sales patterns.

5.3 Feature Engineering

To support business analysis, several new fields were created from the transaction date and time.

The additional fields include:

- **Month Name**
- **Month**
- **Day Name**
- **Day of Week**
- **Hour**

These calculated fields made it possible to analyze customer purchasing behavior across different time periods.

5.4 Data Aggregation

Microsoft Excel Pivot Tables were used to summarize the data across multiple business dimensions, including:

- Revenue by Product Category
- Revenue by Product Type
- Sales by Store Location
- Hourly Sales
- Orders by Day
- Revenue by Product Size

These Pivot Tables formed the foundation of the interactive dashboard.

5.5 Dashboard Development

An interactive dashboard was developed using Microsoft Excel.

The dashboard includes:

- KPI Cards
- Pivot Charts
- Pivot Tables
- Slicers
- Interactive Filters

Users can filter the dashboard by **Month** and **Day**, allowing quick exploration of sales performance from different perspectives.

6. Dashboard Overview

The dashboard was designed to provide management with a clear and interactive overview of business performance.

The dashboard consists of four KPI indicators and six analytical visualizations that answer the project's business questions.

Key Performance Indicators (KPIs)

KPI	Value
Total Sales	\$698,812.33
Total Orders	149,116
Average Price per Order	\$4.69
Average Items per Transaction	1.44

These KPIs provide an immediate snapshot of overall business performance.

The dashboard also includes interactive slicers that allow users to filter the analysis by month and day, making it easier to examine sales trends during specific periods.

7. Dashboard Analysis

7.1 Overall Sales Performance

During the six-month analysis period, the coffee shop generated a total sales revenue of **\$698,812.33** from **149,116 customer orders**.

The average order value was **\$4.69**, while customers purchased an average of **1.44 items per transaction**.

These figures indicate that the business experiences a high volume of customer transactions with relatively low individual purchase values, which is characteristic of the coffee shop industry. Rather than relying on expensive single purchases, the business generates revenue through frequent customer visits and consistent daily sales.

The high number of transactions also suggests a strong customer base and steady demand throughout the analysis period.

7.2 Hourly Sales Analysis

The "Hourly Ordered Quantity" chart illustrates customer purchasing activity throughout the day.

The analysis reveals that customer traffic begins around **6:00 AM** and increases rapidly during the early morning hours. Sales reach their highest level between **8:00 AM and 10:00 AM**, with the peak occurring at approximately **10:00 AM**.

Following the morning rush, sales decrease significantly and stabilize throughout the afternoon before gradually declining during the evening hours.

This trend reflects the typical purchasing behavior of coffee shop customers, many of whom purchase coffee before work or during breakfast hours.

Business Insights

The morning period represents the most critical sales window for the business.

Management should ensure that:

- Additional staff members are scheduled during peak hours.
- High-demand products are fully stocked before opening.
- Coffee machines and equipment are operational before the morning rush begins.

Doing so will improve customer satisfaction while maximizing sales opportunities.

7.3 Revenue by Product Category

The "Revenue by Category" chart shows the percentage contribution of each product category toward total sales revenue.

Category	Revenue Share
Coffee	39%
Loose Tea	28%
Bakery	12%
Drinking Chocolate	10%
Coffee Beans	6%
Branded Products	2%
Flavours	1%
Other	1%

Coffee accounts for nearly forty percent of total revenue, making it the primary source of income for the coffee shop.

Loose Tea is the second-largest contributor, highlighting that non-coffee beverages also play an important role in overall sales performance.

Bakery products contribute an additional twelve percent, demonstrating the importance of complementary food items.

Business Insights

Since coffee is the dominant product category, marketing efforts should continue to focus on premium coffee beverages.

Additionally, bakery products can be bundled with coffee through promotional offers to increase the average transaction value.

7.4 Revenue by Product Size

Product size analysis shows the following distribution:

Size	Revenue Share
Large	34%
Not Defined	31%
Regular	28%
Small	7%

Large-sized beverages generate the highest proportion of revenue.

This indicates that many customers prefer purchasing larger beverages despite the higher price.

However, approximately **31%** of revenue falls under the "Not Defined" category.

Business Insights

The large percentage of undefined product sizes suggests inconsistencies in data entry.

Management should improve point-of-sale data recording by ensuring employees correctly select product sizes during transactions.

Improved data quality will enhance future business analysis and reporting accuracy.

7.5 Store Performance Analysis

Sales performance across the three coffee shop locations is summarized below.

Store	Revenue
Hell's Kitchen	\$236,511.17
Astoria	\$232,243.91
Lower Manhattan	\$230,057.25

Hell's Kitchen generated the highest revenue during the analysis period.

Although it ranks first, the difference between the three stores is relatively small.

Business Insights

The similar revenue generated across all locations indicates consistent operational performance.

Management should compare staffing levels, customer traffic, and promotional strategies across stores to identify the practices contributing to Hell's Kitchen's slight advantage and replicate them at other locations.

7.6 Top Revenue-Generating Products

The dashboard identifies the five highest-performing products.

Product	Revenue
Barista Espresso	\$91,406.20
Brewed Chai Tea	\$77,081.95
Hot Chocolate	\$72,416.00
Gourmet Brewed Coffee	\$70,034.60
Brewed Black Tea	\$47,932.00

Barista Espresso generated the highest revenue among all products.

This demonstrates strong customer demand for premium espresso-based beverages.

Business Insights

The coffee shop should continue promoting Barista Espresso through loyalty programs, featured menu placement, and seasonal offers.

These products should always remain adequately stocked to avoid lost sales.

7.7 Orders by Day of Week

Order counts remain relatively stable throughout the week.

Day	Orders
Friday	21,701
Thursday	21,654
Monday	21,643
Wednesday	21,310
Tuesday	21,202
Sunday	21,096

Saturday **20,510**

Friday records the highest number of orders, while Saturday records the lowest.

The variation between weekdays and weekends is relatively small.

Business Insights

Consistent daily order volumes indicate stable customer demand throughout the week.

Since Fridays experience the highest activity, additional staffing and inventory planning should be considered for that day to maintain service quality.

8. Key Findings

The analysis of the coffee shop sales data provided several valuable insights into business performance, customer behavior, and product trends. The dashboard enabled the identification of key performance indicators and sales patterns that can support strategic decision-making.

The major findings are summarized below:

8.1 Overall Business Performance

- The coffee shop generated a total sales revenue of **\$698,812.33** over the six-month period.
- A total of **149,116 customer orders** were processed.
- The average order value was **\$4.69**.
- Customers purchased an average of **1.44 items per transaction**, indicating that most purchases consisted of one or two products.

8.2 Customer Purchasing Behavior

- Customer activity increased significantly during the morning hours.
- Peak sales occurred between **8:00 AM and 10:00 AM**, making this the busiest period of the day.
- Order volumes remained relatively stable during the afternoon before declining in the evening.

8.3 Product Performance

- Coffee products generated **39%** of the total revenue, making them the most profitable category.
- Loose Tea contributed **28%**, demonstrating strong demand for alternative beverages.
- Bakery products accounted for **12%** of revenue, highlighting their importance as complementary items.

8.4 Product Size Preferences

- Large-sized products generated the highest revenue (**34%**).
- Regular-sized products contributed **28%**.
- Small-sized products generated only **7%** of total revenue.
- A significant portion (**31%**) of sales was categorized as "Not Defined," indicating a need for improved data entry practices.

8.5 Store Performance

- Hell's Kitchen generated the highest revenue (**\$236,511.17**).
- Astoria and Lower Manhattan followed closely behind.
- Revenue differences among the three stores were relatively small, indicating balanced business performance across locations.

8.6 Best-Selling Products

The highest revenue-generating products were:

1. Barista Espresso
2. Brewed Chai Tea
3. Hot Chocolate
4. Gourmet Brewed Coffee
5. Brewed Black Tea

These products represent the strongest contributors to overall revenue and customer demand.

8.7 Weekly Sales Trends

Sales remained relatively consistent throughout the week.

- Friday recorded the highest number of customer orders.
- Saturday recorded the lowest number of orders.

- The difference between weekdays and weekends was minimal, suggesting a stable customer base throughout the week.
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9. Business Recommendations

Based on the findings of this analysis, several recommendations can help improve operational efficiency, customer satisfaction, and overall business performance.

Recommendation 1: Optimize Staffing During Peak Hours

The analysis indicates that the highest customer traffic occurs between **8:00 AM and 10:00 AM**.

Management should schedule additional employees during these hours to:

- Reduce customer waiting times.
- Improve service efficiency.
- Increase customer satisfaction.

Adequate staffing during peak periods will help maximize revenue while maintaining service quality.

Recommendation 2: Focus Marketing on High-Revenue Products

Barista Espresso is the highest revenue-generating product.

The coffee shop should:

- Highlight this product in promotional campaigns.
- Feature it prominently on menus.
- Include it in loyalty programs and seasonal promotions.

Promoting best-selling products can further increase revenue.

Recommendation 3: Increase Cross-Selling Opportunities

Coffee and bakery products complement each other naturally.

The business can introduce promotional bundles such as:

- Coffee + Croissant
- Espresso + Muffin
- Tea + Cookie

These offers can encourage customers to purchase additional items and increase the average transaction value.

Recommendation 4: Improve Data Quality

Approximately **31%** of product size records are categorized as "Not Defined."

This limits the accuracy of future analyses.

Management should:

- Ensure staff consistently record product sizes.
- Validate data entered into the point-of-sale system.
- Standardize data entry procedures.

Better data quality will improve reporting and support more informed decision-making.

Recommendation 5: Monitor Store Performance Regularly

Although Hell's Kitchen currently performs slightly better than the other locations, the difference is relatively small.

Management should:

- Compare operational practices across stores.
- Share successful strategies between branches.
- Monitor monthly performance using the dashboard.

This approach can help maintain consistent performance across all locations.

Recommendation 6: Introduce Customer Loyalty Programs

Since customers typically purchase only **1.44 items per transaction**, encouraging repeat purchases can increase overall revenue.

Possible initiatives include:

- Digital loyalty cards.
- Points-based rewards.
- Buy-one-get-one offers.
- Personalized promotions for frequent customers.

These programs can strengthen customer retention and increase sales.

Recommendation 7: Expand Successful Product Lines

Coffee and tea products account for the majority of total revenue.

Management should consider:

- Introducing new coffee flavors.
- Offering seasonal beverages.
- Expanding premium coffee options.
- Testing limited-time menu items.

Expanding successful categories can attract new customers while encouraging existing customers to try additional products.

10. Conclusion

This project successfully demonstrates how Microsoft Excel can be used as a powerful business intelligence tool for analyzing retail sales data. By applying data cleaning techniques, feature engineering, Pivot Tables, Pivot Charts, and interactive dashboard design, raw transactional data was transformed into meaningful insights that support business decision-making.

The dashboard provides a comprehensive overview of the coffee shop's sales performance, enabling management to evaluate key performance indicators, monitor product performance,

compare store locations, identify customer purchasing trends, and analyze sales patterns throughout the day and week.

The analysis revealed that coffee products are the primary revenue driver, with Barista Espresso emerging as the highest-performing product. Customer demand is strongest during the morning hours, particularly between **8:00 AM and 10:00 AM**, emphasizing the importance of efficient staffing and inventory planning during peak periods. Store performance remained relatively balanced across all locations, while the product size analysis highlighted opportunities to improve data quality by reducing undefined entries.

Overall, this project demonstrates the value of data analytics in supporting strategic business decisions. The insights generated through the dashboard can help management optimize operations, improve customer satisfaction, and increase revenue through informed planning and targeted business strategies.